

Bi-Weekly Status Report 3

Project Name: CodeTrail – Real-Time Collaborative Code Editor with Contribution Tracking
Course: WS & SOA (IT645) | Dhirubhai Ambani University (Formerly DA-IICT)
Report Period: 01/09/2026 – 13/09/2026 (Weeks 6–7)
Date Submitted: 17/09/2026
Prepared By: Shaikh Maryam on behalf of CodeTrail Team

Project Status Summary¹

Sprint / Milestone	SCRUM Sprint 3 (Bi-Weekly 3) – Module 3 Delivery: Real-Time Collaborative Editor			
% Complete (overall)	37.5% (3 of 8 functional modules delivered On Track)			
Indicator	Current	Previous	Trend	Status
Overall Project	37.5%	25%	↑	On Track
Schedule	On Schedule	On Schedule	→	On Track
Scope	3/8 Modules	2/8 Modules	↑	On Track
Quality	No critical defects; viewer guard & version conflict resolution in place	Stable	→	On Track
Team Contribution	8/8 members contributed	8/8 members contributed	→	On Track
Dependencies	Socket.IO integrated; Piston API client 60% complete	Piston API design only	↑	On Track

Summary Notes:
The CodeTrail team successfully completed and merged Module 3 (Real-Time Collaborative Code Editor) in Bi-Weekly 3 (01/09/2026–13/09/2026, Weeks 6–7), delivering two major milestone features: full Monaco Editor integration with live multi-file management, and a production-grade real-time collaboration infrastructure built on Socket.IO. The period also saw the completion of the workspace session management layer (REST APIs, Socket event handlers, and a dedicated React hook), role-based access control enforced at both the frontend and backend socket layer, and a viewer permission guard preventing unauthorized edits. A significant platform-stability milestone was reached with the implementation of monotonic version counters and packet-conflict resolution (CT-89), which systematically resolves concurrent editing

¹For details on indicators and status levels, see last page.

race conditions identified as Risk R-02 in Bi-Weekly 2. The team is fully on schedule with three of eight modules delivered, and Piston API integration (Module 4) is actively underway.

Key Accomplishments in Two Weeks

- **Module 3 Delivery (Target Bi-Weekly Deliverable) – Monaco Editor & Real-Time Collaboration:** Fully integrated `@monaco-editor/react` with multi-file tab switching, language-aware syntax highlighting (JS, TS, PY, C++, Java, Rust, Go, HTML, CSS, JSON, Markdown, SQL, YAML), dynamic file tree with create/delete/rename support, and localStorage-backed content persistence per workspace session. (Commits 99a65acd, 9b8afc5a; PR #15, #17)
- **Real-Time Socket.IO Collaboration Infrastructure (CT-85):** Built dual-layer Socket.IO architecture with `workspaceSessionSocket.js` and `socketHandler.js`, providing room-based code broadcasting (`code_change/code_updated`), heartbeat-driven session time accumulation (`session_heartbeat`), live presence updates (`presenceUpdate`), and graceful disconnect/reconnect handling. (Commit bd7a6894; PR #16)
- **Real-Time Multiplayer Monaco Cursor Tracking & Decorations (CT-86):** Implemented peer cursor rendering with color-coded nametags injected as Monaco delta-decorations; throttled `cursor_position_update` broadcast at 30ms to prevent network flooding; peer selections rendered as translucent color overlays per user. (Commit 25c2b0f6; PR #17)
- **Viewer Role Lock & Version Conflict Resolution (CT-89):** Enforced viewer-read-only guard at both the React component layer and the backend Socket.IO handler; implemented monotonic version counters per file key (`workspaceId:fileId`) to detect and silently drop stale/conflicting code packets; added real-time reconnection status banner with animated indicator. (Commit 9b8afc5a; PR #17)
- **Workspace Session Management Layer:** Delivered `WorkspaceSession` Mongoose model with compound unique index, REST session APIs (enter, leave, heartbeat, file-focus, all-sessions), `workspaceSessionController.js`, session routes, and `useWorkspaceSession.js` custom hook encapsulating Socket.IO lifecycle, code/cursor emission, and version-state management. (Commit a75843fa; PR #14)
- **Workspace Settings Modal & Role-Based UI:** Completed `WorkspaceSettingsModal.jsx` (1,500+ lines) covering member management (invite by email, role promotion/demotion, remove), workspace metadata editing (title, description, language, brand color, icon, public/private toggle), activity history log panel, and session time display – all conditioned on the authenticated user’s workspace role. (Commit a75843fa; PR #14)
- **2.5-Second Debounced Auto-Save (CT-88):** Implemented debounced PATCH request to `/api/workspaces/:id` that persists all file contents to MongoDB Atlas; falls back gracefully to localStorage on network failure with “Saved locally (offline)” status indicator. (Commit a75843fa; PR #14)

Planned vs. Actual

Task/ Deliverable	Planned Completion	Current Forecast	Actual Status	Notes
Requirements Eng. (BRD / RSD)	Aug 24	—	Done	Submitted PR #12 (Bi-Weekly 2)
System Design (SDD)	Aug 31	—	Done	Authored by Meet (Bi-Weekly 2)
Module 1: Auth & User Mgt	Aug 17	—	Done	Submitted in Bi-Weekly 1
Module 2: Workspace Mgt	Aug 31	—	Done	Merged to main (Bi-Weekly 2)
Module 3: Collaborative Editor	Sep 14	—	Done	Monaco + Socket.IO delivered; merged PR #15, #16, #17
Module 4: Code Execution	Sep 28	Sep 28	In Progress	Piston API client ~60% complete
SHA-256 Contribution Logging	Sep 28	Oct 05	In Progress	Core pipeline design underway (Bi-Weekly 4 target)
Module 5: Contribution Dossier	Oct 12	Oct 12	Not Started	Scheduled for Bi-Weekly 5

Project Risks, Issues, & Dependencies

Type (ID)	Description	Impact	Owner	Mitigation / Next Step
Risk (R-02)	Concurrent typing race conditions during live collaborative editing	High	Mahi Kansara / Anistina Dsouza	Resolved (CT-89): Monotonic version counters and stale-packet dropping implemented and merged.
Risk (R-03)	Monaco Editor memory growth from unbounded peer cursor decorations	Medium	Dudhat Monar	Delta-decoration IDs tracked in <code>remoteDecorationIdsRef</code> ; old decorations replaced atomically. Monitor under load.
Issue (I-02)	Concurrency collision on atomic participant additions via invite codes	Medium	Manushri Thakkar / Ruchita Patadiya	Resolved in Bi-Weekly 2; <code>\$addToSet</code> and unique indexing in place.
Dependency (D-02)	Third-party Piston API execution rate limits and latency variability	Medium	Meet Sheth / Dudhat Monar	Execution timeout guards and client debouncing designed; Piston client wrapper 60% complete (CT-450).
Risk (R-04)	Socket.IO connection flapping under high-load / network instability	Low–Medium	Manushri Thakkar	10-attempt auto-reconnect, exponential back-off, and reconnection UI banner in place.

Project Decisions

Project Scope Changes

Upcoming Work (Next Two Weeks' Plan – Bi-Weekly 4 Target)

- **Deliver Module 4: Code Execution Engine** – Complete Piston API client service wrapper (CT-450); integrate language runtime execution (JavaScript, Python, C++, Java, Go, Rust) into the live workspace; display real-time stdout/stderr output in the Terminal panel.
- Establish the core SHA-256 cryptographic activity logging pipeline for tamper-evident per-user contribution records (CT-460).
- Implement dynamic file delete and rename operations within live workspace sessions, with Socket.IO broadcast to peers.
- Begin Module 5 architecture: Contribution Dossier (aggregation of per-user code diff metrics and session durations).

Decision	Rationale	Impact	Decided By	Date
Dual-layer Socket.IO architecture (<code>socketHandler.js</code> + <code>workspaceSessionSocketHandler.js</code>)	Separates in-memory real-time presence from DB-backed session persistence for clearer responsibility	Cleaner architecture; easier to test each layer independently	Team	Sep. 4
Monotonic version counters per <code>workspaceId:fileId</code> key	Eliminates packet-order ambiguity in multi-client concurrent edits without requiring OT or CRDT overhead at this stage	Prevents data corruption from stale broadcast packets (CT-89 resolved)	Mahi / Manushri	Sep. 10
30 ms cursor throttle with pending-timeout pattern	Prevents cursor event flooding on fast typists (>30 events/sec) while maintaining sub-frame-perceptible latency	Reduced socket bandwidth; smoother peer cursor UX (CT-86)	Dudhat Monar	Sep. 13
localStorage fallback for auto-save	Ensures work is not lost if backend is temporarily unreachable during development/testing	Increased resilience; “Saved locally (offline)” graceful degradation	Maryam Shaikh	Sep. 9
Viewer role enforced at socket layer (not just UI)	Prevents malicious or accidental code changes from clients that bypass the frontend guard	Security hardening of collaborative session (CT-89)	Mahi Kansara	Sep. 13

Change	Action	Reason	Schedule Impact	Approved By
None	—	Scope remains fully aligned with project proposal and approved SDD specifications.	None	—

- Expand Postman API test suite to cover Socket.IO event contracts and execution engine endpoints.

Team Member Contributions

Name: Meet Sheth (202512057)

Role: Backend Architect / System Designer

Tasks Completed	Refactored <code>WorkspaceModal.jsx</code> with expanded create-and-join flows and improved validation UX. Updated <code>HomePage.jsx</code> with quick-join action, workspace language tag normalization, and active filter styling. Extended <code>index.css</code> with modal and dashboard layout refinements.
Tasks In Progress	CT-425 (MongoDB query optimization – indexing strategy for workspace and session collections) CT-420 (Socket.IO workspace room architecture review and hardening)
Contribution Type	Development / Architecture
Blockers	None
Hours / Capacity	26/30 hrs logged
Evidence	<code>WorkspaceModal.jsx</code> , <code>HomePage.jsx</code> , <code>index.css</code>

Name: Rishika Shah (202512041)

Role: Frontend Developer / UI Engineer

Tasks Completed	Refactored <code>WorkspaceModal.jsx</code> (+109 lines): expanded create-and-join flows, improved validation UX. Updated <code>HomePage.jsx</code> (+61 lines): quick-join action, language tag normalization, active filter glow styling. Extended <code>index.css</code> (+65 lines): modal and dashboard layout refinements. (Commit <code>f16eff1b</code> , Sep. 1; merged via branch <code>features/Rishika</code>)
Tasks In Progress	CT-455 (Collaborative editor test scenarios and validation scripts for FR-07 to FR-12) – 65% complete; expected finish Sep. 28.
Contribution Type	Development / Testing
Blockers	None
Hours / Capacity	30/30 hrs logged
Evidence	Commit <code>f16eff1b</code> (Sep. 1); <code>WorkspaceModal.jsx</code> , <code>HomePage.jsx</code> , <code>index.css</code>

Name: Mahi Kansara (202512078)

Role: Frontend Developer / Socket.IO Integration Lead

Tasks Completed	Phase 3B (CT-89), 6 files, 468 insertions (Commit 9b8afc5a, Sep. 13, PR #17): ModularWorkspace.jsx (+282): viewer isViewer guard, reconnection banner, fileVersions/fileVersionsRef sync, stale-packet drop. useWorkspaceSession.js (+75): code_conflict handler, code_change_error listener. workspaceSessionSocket.js (+72): viewer code_change block, version-counter, code_conflict emit. socketHandler.js (+60): same viewer guard and version-counter on second socket layer. auth.js (+15): middleware hardening. Workspace.js (+1): model field addition.
Tasks In Progress	CT-445 (Monaco Editor & Socket.IO synchronization architecture refinements) – 90% complete.
Contribution Type	Development / Architecture
Blockers	None
Hours / Capacity	30/30 hrs logged
Evidence	Commit 9b8afc5a (Sep. 13); PR #17; ModularWorkspace.jsx, useWorkspaceSession.js, workspaceSessionSocket.js, socketHandler.js, auth.js, Workspace.js

Name: Thakkar Manushri (202512122)

Role: Project Manager / DevOps / Backend Developer

Tasks Completed	Commit bd7a6894 “added sockets” (PR #16, Sep. 12, 318 insertions, 5 files): workspaceController.js (+54): ObjectId.isValid() guards on 6 endpoints (getById, update, delete, session, history GET/POST). workspaceSessionSocket.js (+19): initial code_change broadcast handler (CT-85 first pass). ModularWorkspace.jsx (+244): role resolution refactor (isOwner, memberObj), filesContentRef & autoSaveTimerRef hooks. Also merged PRs #14, #15, #16, #17 and coordinated Sprint 3 Jira backlog.
Tasks In Progress	CT-420 (Socket.IO room architecture hardening) – 75% complete. CT-425 (MongoDB query optimization) – 50% complete.
Contribution Type	Development / Project Management / DevOps
Blockers	None
Hours / Capacity	28/30 hrs logged
Evidence	Commit bd7a6894 (Sep. 12); PR #16; merge approvals (#14, #15, #17); workspaceController.js, workspaceSessionSocket.js, ModularWorkspace.jsx; Jira CT-325

Name: Anistina Dsouza (202512081)

Role: Full-Stack Developer / Module Lead

Tasks Completed	No direct code commit this period (contributions carried from Bi-Weekly 2). Collaborated on review and testing of the <code>ModularWorkspace.jsx</code> multi-file management feature; supported Ruchita Patadiya's shared editor implementation (PR #15); continued work on CT-435 workspace participant role permissions.
Tasks In Progress	CT-435 (Workspace participant role permissions and member management API) – 90% complete; expected to close in Bi-Weekly 4.
Contribution Type	Testing / Project Management
Blockers	None
Hours / Capacity	24/30 hrs logged
Evidence	CT-435 Jira subtask; code review on PR #15 and PR #16

Name: Ruchita Patadiya (202512070)

Role: Frontend Developer / Backend Developer

Tasks Completed	Commit 99a65acd “shared code editor and multi-file management” (PR #15, Sep. 12, 254 insertions): <code>ModularWorkspace.jsx</code> (+258 net, –74 refactor): integrated <code>@monaco-editor/react</code> ; authored <code>getMonacoLanguage()</code> (16 file types) and <code>getStarterBoilerplate()</code> per-extension generator; defined <code>DEFAULT_WORKSPACE_FILES</code> and <code>DEFAULT_FILES_CONTENT</code> ; refactored <code>customFiles/filesContent</code> state with <code>localStorage</code> backup and <code>activeWorkspace.files</code> overlay. <code>package.json</code> (+1): added <code>@monaco-editor/react</code> dependency.
Tasks In Progress	CT-430 (Workspace authorization and role-based route guards) – 80% complete.
Contribution Type	Development
Blockers	None
Hours / Capacity	30/30 hrs logged
Evidence	Commit 99a65acd (Sep. 12); PR #15; <code>ModularWorkspace.jsx</code> , <code>frontend/package.json</code>

Name: Shaikh Maryam (202512074)

Role: Full-Stack Developer / UI Architect

Tasks Completed	Commits a75843fa (PR #14, Sep. 9), 06854877 (Sep. 7), 0bf8a7b0 (Sep. 4) – 6,424 insertions across 23 files: NEW files: workspaceSessionController.js (225 lines, 5 REST APIs: enter/get/all-sessions/heartbeat/file-focus/leave); WorkspaceSession.js (54 lines, Mongoose model, compound unique index); workspaceSessionRoutes.js; WorkspaceSettingsModal.jsx (1,303 lines: member invite, role management, history log, metadata edit); WorkspacesPage.jsx (610 lines: public workspace discovery). Extended: workspaceController.js (+550), socketHandler.js (+155), workspaceSessionSocket.js (+163), ModularWorkspace.jsx (+1,314), index.css (+1,596), HomePage.jsx (+356), App.jsx (+77), Navbar.jsx (+66).
Tasks In Progress	CT-478 (Collaborative editor layout and multi-panel split-view) – 80% complete.
Contribution Type	Development / Architecture / Documentation
Blockers	None
Hours / Capacity	30/30 hrs logged
Evidence	Commits a75843fa, 06854877, 0bf8a7b0; PR #14; WorkspaceSession.js, workspaceSessionController.js, WorkspaceSettingsModal.jsx, WorkspacesPage.jsx, ModularWorkspace.jsx, index.css

Name: Dudhat Monar (202512019)

Role: Backend Developer / QA Engineer

Tasks Completed	Commit 25c2b0f6 – CT-86 multiplayer cursor tracking (PR #17, Sep. 13, 425 insertions, 5 files): workspaceSessionSocket.js (+40): cursor_position_update & cursor_move handlers broadcasting cursor_position_updated/cursor_updated to room peers. socketHandler.js (+21): same cursor handlers on second socket layer. useWorkspaceSession.js (+45): cursor_position_updated listener updating peerCursors; emitCursorPosition() function. ModularWorkspace.jsx (+259): PEER_COLORS, getPeerColor(), updateMonacoDecorations() with dynamic CSS nametag injection and delta-decoration API; 30 ms throttle. index.css (+66): .ct-monaco-remote-cursor, .ct-monaco-nametag, .ct-monaco-remote-selection rules. Also expanded Postman API test suite for Session REST endpoints.
Tasks In Progress	CT-440 (Automated API integration test suite for workspace and session routes) – 70% complete.
Contribution Type	Development / Testing
Blockers	None
Hours / Capacity	28/30 hrs logged
Evidence	Commit 25c2b0f6 (Sep. 13); PR #17; workspaceSessionSocket.js, socketHandler.js, useWorkspaceSession.js, ModularWorkspace.jsx, index.css; Postman test collection

Tip: If your team uses Jira, you can fill this section by pulling closed/in-progress tickets assigned to each person, rather than filling it manually.

Guidelines (for reference, not to be submitted in your status report):

- Section 1 summarizes the overall project status.
- Sections 2–5 capture team-wide progress.
- Section 6 is the per-person breakdown – keep individual entries short (3–5 bullet points max) so the report does not balloon as your project progresses / team grows.
- The status indicators at the top give stakeholders a one-glance read before they dig into details – useful if this report goes to non-technical stakeholders too.
- Six project status indicators have the following context:
 - Overall Progress — % of project scope completed
 - Schedule — % of planned work completed
 - Scope — % of approved requirements completed
 - Quality — test pass rate / critical defects
 - Team Contribution — % of assigned deliverables completed
 - Dependencies — % of critical dependencies resolved
- Status levels:
 - On Track — expected to meet current milestone
 - Needs Attention — issue exists but recovery is likely
 - At Risk — milestone/deliverable may be missed without intervention
 - Blocked — progress cannot continue without resolving a dependency
- Contribution Type with examples (Add more types, if needed)
 - Development — Code, APIs, UI, database
 - Architecture — Architecture/design decisions
 - Testing — Test cases, automation, defect verification
 - Documentation — SRS, design docs, reports
 - DevOps — CI/CD, deployment, containers
 - Research — Literature review, technical investigation
 - Project Management — Planning, coordination
 - Presentation — Demo, presentation, client communication